

5. Modulation of autoISF aggressiveness.

V 4.6

Please note that with autoISF you are in an early-dev. environment, where the user interface is **not optimized for safety** of users who stray away from intended ways to use. Good safety features exist, but these are only as good as the development-oriented user understands and implements them. This is not a medical product. Refer to disclaimer in [section 0/readme.md](#)



5.1 Automatic modulation of loop aggressiveness

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Summary: Your personal FCL cockpit

[Available \(related\) case studies:](#)

Case study 5.2: Sweet snack.

Case study 5.3: Compression low

Once the initial tuning according to [section 4.](#) is done, you are ready to use autoISF for your fully automated *meal management*.

Challenges beyond managing main meals

There are up to four *other major challenges* you might have to manage:

1. Secure the technical pre-requisites for your FCL remain a given (see [section 1](#)):
 - Prevent, recognize and manage (partial) occlusions (=>frequent cannula changes)
 - Secure “meaningfulness” of 100% all your CGM values

- Secure 24/7 BT connectivity of all your components

2. Deal with times when insulin given, after meal detection, must be restricted, e.g.:

- a snack should not be “treated” as a meal
- meal before exercise should not be treated “all the way” towards high iob.

3. Deal with times when the loop should be set “less aggressive” as a precaution, e.g.

- during nights
- in an exercise context

4. Deal with times when the loop should be set “more aggressive” e.g.

- at sickness
- at tempting hotel breakfast buffet, or other “diet sin” + in-activity

How big the remaining challenge really is, depends very much on your targeted %TIR and:

- a) on your **individual lifestyle**: => is your loop (automatically, [section 4.1](#), or via a needed intervention, [section 4.2](#)) prepared to respond with required different aggressiveness?
- b) on **quality of your tuning**: => will your loop frequently “drift outside of the desired corridor”? Will it automatically recognize this, and pull back into it? Can you recognize it, and pull back earlier?

Hands-off FCL – an elusive goal?

To run the FCL “hands-off” around the clock, you now must analyze **your** data also in the times *outside* the meal blocks, and seek solutions to problems (if any).

With a technically well-functioning system, *moderate meals*, *moderate exercise*, moderate %TIR expectations and a bit of mindfulness it should be possible to go into **Full Closed Loop 24/7**, after working through, and observing, [sections 1-4](#). See [case study 4.3](#).

- In [section 5.1](#) we explore avenues towards **fully automated** management of *some other “disturbances”* that, in daily life, also will not require any user intervention.

Often it will be **your** choice whether you want to bother with researching in **your** data for, and defining, dedicated (“personalized”) Automations (see also [section 5.1.6.](#)).

Or whether you accept instances where you do manual steps:

- In [section 5.2](#) (and later, related to exercise, in [section 6.](#)) we will look at solutions that involve an easy **user interaction** like a data entry, or a button push.

Avenues for temporary (automatic or manual) modulation of the FCL aggressiveness

1. temporary shut-off SMBs (via odd -numbered target)
2. temporary change ISF-weights
3. temporary change iobTH_percent
4. temporary change the set %profile
5. temporary set different bg target (especially in connection with exercise mode)t

All of these are also easy accessible via Automations in AAPS

This means: If, in **your** data, you find sets of conditions where any of the actions 1.-5. could help, you can make this aspect integrated part of **your** hands-off FCL!

5.1 Fully automatic modulation of FCL aggressiveness

The following subchapters describe set-ups you may want to use for allowing **completely hands-off FCL** in as many daily situations as possible (and potentially all the time, as in [case study 4.3](#))

Note that in Automations you can go beyond above mentioned 5 avenues, and come up with a nearly surgical definition as to from which iob on, and many other criteria, some temporary modulation of aggressiveness shall happen.

5.1.1 All autoISF ISF adaptations switched off outside of meal-time windows

If, aside from having to bolus for meals, your hybrid closed loop was running pretty well *without* other interventions from your side, you could continue to run in that mode, and just focus your new autoISF FCL on management of meals.

In your initial transitioning phase, this approach makes a lot of sense, and even by focusing autoISF on just a sub-set of meals, like only dinners.

Also in the long run this avenue is taken by many FCL users for the night times, “hanging on” to their well performing hybrid closed loop with standard oref(1) SMB+UAM

To install this mode, you define Automations

- that set meal time windows in which “Enable ISF adaptation by glucose behavior” (autoISF) is turned on in AAPS preferences/OpenAPS SMB
- or: that turn *all* autoISF's ISF modulations (or just *bgAccel_ISF*) off in time windows in which surely no meal occurs.

For instance, you can go *for all nights* back into your Hybrid Closed Loop, as you had before.

Your temp. “autoISF shut-down” (exiting autoISF FCL = shutting off “Enable ISF adaptation by glucose behavior” in AAPS Preferences / SMB) is meant to prevent problems from the loop *over-reacting* to bumps in the glucose curve in times of day (night), when standard oref(1) performance is sufficient.

A very good alternative to defining “meal-time windows”, or to resorting to night-time Hybrid Closed Loop, is letting the autoISF FCL run 24/7, and “taming” the FCL via a night time SMB shut-off (see next [section 5.1.2](#)).

Off topic: **Other “early dev” AAPS variants** (see [section 13.3](#)), do require working with meal-time windows. The window is either set by time of day in the settings, or it always must be „set“ by the user. Trigger to set a meal time window could be a pre-bolus given by the user, a carb entry made, an EatingSoonTT set, or a meal-announcement-button pushed (none of these things are required in autoISF FCL). Outside of these time windows, these loops then run with less aggressive SMBs, just like oref(1) SMB+UAM in AAPS Master, with or without (modified versions of) dynamicISF. So, outside of the meal time windows, you would be in “your normal” hybrid closed loop. The term **Meal Announcement** (MA) is often used to label this closed looping mode. It is not really FCL, but an advance over traditional HCL.

5.1.2 Odd-numbered **profile targets**, to block SMBs

An alternative route of preventing the FCL loop from over-reacting to bumps in the glucose curve would be to make use of the option to temporarily shut down SMBs

Ensure the even/odd logic in the settings is toggled on in Preferences> openAPS SMB> autoISF settings> smb delivery settings>: "Enable alternative activation of SMB depending on bg target": ON.

154

In time blocks with an odd-numbered profile target you can prevent any SMBs being given by your loop. The (unchanged) aggressive settings then can only translate within the limits set by %TBR possible.

158

This will very much slow down any more insulin being given, and is an excellent solution for night times, especially if you occasionally experience compression lows.

161

Alternatively, you could use the new included options for Automation Conditions and

- allow/block SMBs via Automation commands, or:

- temporarily tune your bgAccel_ISF_weight much lower (see [section 5.1.4](#)).

165

The same situation can be achieved if you generally operate with a mild bgAccel_ISF, and make your autoISF only really aggressive for meal-time slots (if you have similar enough times every day, or also can “employ” geo-fencing in your Automation (or middleware, in iAPS) conditions).

In these cases you would not need to have night profiles that disable SMBs: - Which is the better way would depend on a lot of personal factors relating to how high-carb the diet is, regularity of meals, snacking habit, CGM quality and incidence of compression lows, and probably more. - I would try both routes, or, as this is fairly complex to tune, just one, and stick with what is working good enough.

Yet another alternative was already presented ([section 5.1.1](#)) = to go into hybrid closed loop for the night.

That is possible, with SMBs available (without them getting boosted via autoISF), and, for a long time, was the author’s favored solution for the nights.

This solution is evidently similar to the prior discussed one, of having a mildly tuned autoISF 24/7, boosted to high aggressiveness only in meal-time slots.

180

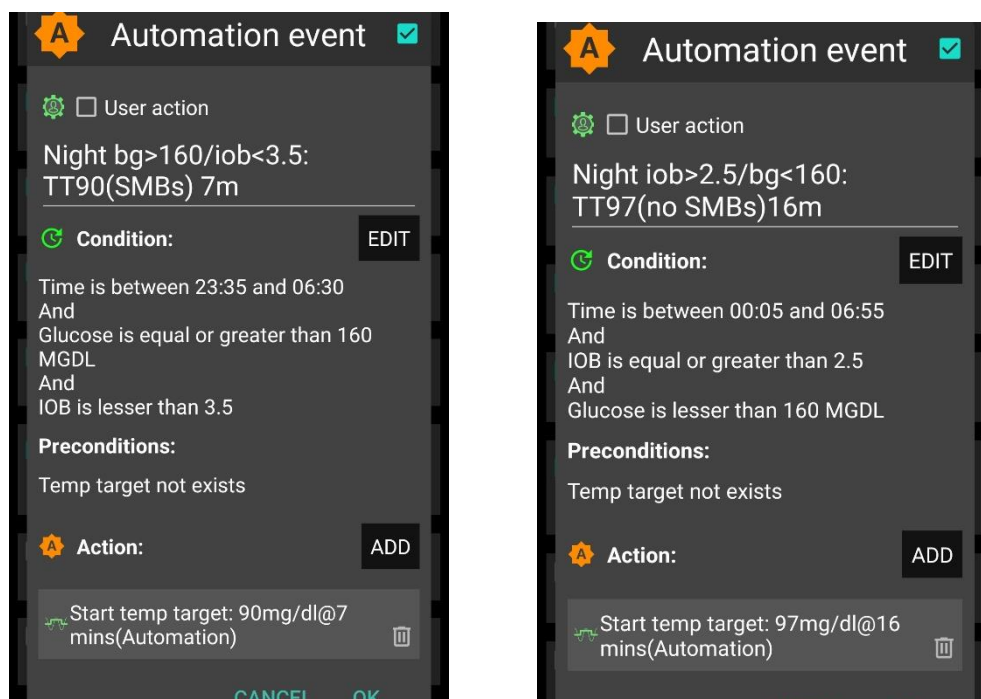
[Adjunct Automations to allow a few SMBs, in nights with odd profile target SMB shut-out](#)

182

My **favorite** builds on the **method** of this section (5.1.2, odd **profile** target provides SMB shut-out at night), but then allowing some, automatically triggered SMBs, when needed:

185

186 In case you occasionally do have nights that would benefit from a couple of SMBs (to treat temp.
 187 highs from a late fatty pizza, raclette and such): Define suitable Automations like the two „night“
 188 ones in this *example*:



189
190

191 Caution: Never underestimate the „trickiness“ of getting your Automations „right“.

192

193 With your thought-out Automations in place, night data need to be analyzed to see

- 194 • whether the bg and iob limits, as defined in the given example, work sensibly four your data
- 195 pattern
- 196 • whether the TT duration is chosen appropriately
- 197 • how swapping the sequence in which the automations appear in the Automation list would
- 198 lead to different SMB impacts.

199

200 5.1.3 Odd-numbered **temp. targets (TT)** set via Automation, to block SMBs

201

202 A widely used Action that strongly modifies how fast your FCL can add more iob is setting an **odd-**
 203 numbered **temp. glucose target** which makes the loop operate without giving any SMBs (%TBR
 204 modulation only).

205 Ensure the even/odd logic in the settings is toggled on in Preferences> openAPS SMB>
 206 autoISF settings> smb delivery settings>: "Enable alternative activation of SMB depending
 207 on bg target": ON.

208

209 Then, from patterns you find in *your* data, at times where you want your loop act differently, you
210 need to carve out Conditions that describe the respective situations (and either *for how long* it
211 typically lasts, or at which *other* Conditions you want your loop get back to default FCL operation).

212

213 An odd TT is often set for an *anti-hypo* snack or *sports* snack. In both instances, you do not want
214 SMBs to quickly counter-act.

215

216 Note that in case of *sweet “fun”* snacks, this is entirely different -> [section, 5.2.1](#) or for
217 regular snacks (*e.g. at school break*) see next [section 5.1.4](#)

218

219 5.1.4 Automatic differentiation of FCL aggressiveness using Automations (or middleware)

220

221 **Personalized Automations** tailor the loop exactly to ***your*** data so **fully automated** handling of
222 situations with **different aggressiveness** of the loop can be made.

223

224 Automations are an integrated and very easy-to-use feature in AAPS.

225 (The i-Phone platforms Trio or iAPS lack this feature. However, so-called **middleware** has been
226 developed as add-in to your code, see: <https://github.com/macconnellk/RoboSurfer/tree/main>)

227 .

228 From, autoISF 3.0 onwards, also the following parameters are provided as Condition and/or as
229 Action for defining YOUR Automations:

- 230 • Enable ISF adaptations by glucose behavior => Allows temp. ON/OFF for the key ISF
231 modulation parts of autoISF (and, as a result, will usually decrease loop aggressiveness)
- 232 • Trigger/set iobTH percent => Keeps default aggressiveness, but only until a iob threshold
233 (that your Automation modifies) is surpassed (which is when any further SMBs will be
234 blocked blocked)
- 235 • Trigger/set bgAccel_ISF_weight => Modifies the aggressiveness of just the acceleration
236 component

237 To set up suitable Automations, you first must **analyze patterns** you find **in your data**, at times (or
238 geo-location, or bg and iob patterns that point to a problem ...) **where you want your loop act**
239 **differently**, to carve out Conditions that describe the respective situations (and either for how long
240 it typically lasts, or at which *other* Conditions you want your loop get back to default FCL
241 operation).

242

243 A variant of this mode is to define several windows in which autoISF aggressiveness
244 (bgAccel_ISF_weight) and/or iobTH are automatically set differently

- 245 • for **different meal time slots** of your day –

(Breakfast at home, school lunches, school intermission snacks, dinners at home could for example all deserve special settings regarding ISF_weights and iobTH).

Note: Circadian differences in insulin sensitivity between meal times are included via your ISF profile and should not be a reason for different _weights needed between meals!

- or even for a geo-location etc –

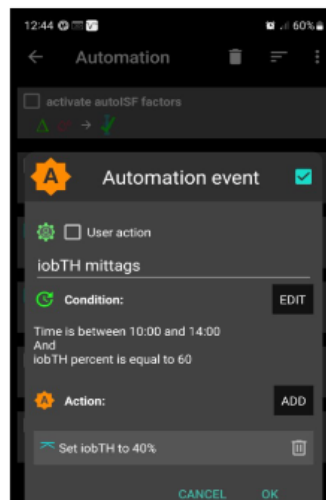
(School lunches, or mother-in-law visits, would be examples).

An example for this was given in section 3 already:

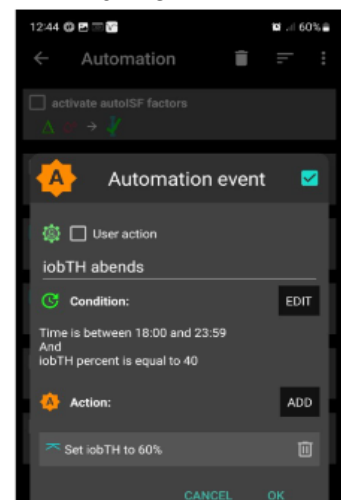
Here is an example set of automations to alternate between two values of iobTH:

I use two different values of *iob_threshold_percent* during a normal day. It is 40% for lunch time and 60% for dinner time. I have these two rules to switch by time of day and only if the current value equals the value from the earlier shift. Any other value is treated as a manual override for special occasions until I manually set it to its regular value. The time windows for switching are long enough to catch an opportunity to be processed and do not need to be actioned half a day each.

lunch



dinner



Unless your meals differ vastly in size and in fast carb content **all this may not be needed.**

- Note: Only the main two parameters (bgAccel_ISF_weight for “initial aggressiveness”, and iobTH_percent for “where SMBs stop”) are available in Automations. So, finding your parameter sets *for each of the time slots*, will not be trivial. => **Spending more effort to set the .._weights so they accommodate just one, broader spectrum (section 4.) should be the first, and standard, approach.**
- An intermediate (maybe only temporary) approach could be to use a **profile switch** (for low carb meal, or eating half, setting %profile to 60% for instance, and only for the brief, less than an hour, initial meal period). See [section 5.2.2](#)

Still, personalized Automations might help ease your initial job of setting the various ISF_weights, and of finding a best-suitable iob_threshold_percent that would work “always”.

Note: Setting a different iobTH% or bgAccel_ISF_weight can probably not be done with a duration attached. Then you **must** define a suitable **additional Automation** (example see next page) that must be active in tandem, to **restore the values you had set in /Preferences for your iobTH% or bgAccel-ISF_weight.**

Else, once your Automation set in, it will *forever* shift these important parameter settings!

272

273 If for instance you have several Automations that, in
274 combination with a set elevated TT also set a lower iobTH:
275 Don't be fooled, the duration only applies to the TT.

276

277 You need an extra Automation for all of them.

278

279 I picked out the highest of the altered iobTH values that these
280 Automations can set (*in my case, 45% or lower*), and then I can
281 automatically restore *my default desired 60%, via this one* - ->

281

282

283 **Caution:** Resist the temptation to use too many Automations. Notably, do not include Automations
284 that would duplicate (and in effect delay) the job that autoISF tries to do:

285 Automations: Relative **disadvantages** to/when using autoISF ([Discord 26.8.2025/ chiakii/ qa-zelle](#)):

286 “..I've made an **automation** that sets a low TT on high bg deltas, but when it gets triggered, the SMB I get is calculated
287 with the "before TT" weights/values, and only the next tick (next CGM value) will take the low TT into account.
288 Given that we want to front load the SMBs as much as possible, that's **5 minutes lost** with a high delta, which isn't
289 ideal”

290 ANSWER: A new bg reading triggers a new loop and checking the automation triggers. Therefore it comes too late.
291 Your scenario reminds me of cases with too **many automations** that **interfere with the** (autoISF) **built in control**
292 **logic!** Note that if you observe large delta, then inevitably there was a high acceleration **before**. In terms of front
293 loading that **acceleration** could have done its job when you use **autoISF**. Even a stronger **pp ISF** weight would
294 probably help more.

295

296 5.1.5 Automatic adjustment of FCL aggressiveness via the Activity Monitor

297

298 With the autoISF variant of AAPS you can make use of your smartphone's **stepcounter** and use it
299 to fully automatically adjust insulin sensitivity ratio to **activity level in the past minutes to one**
300 **hour** time frame.

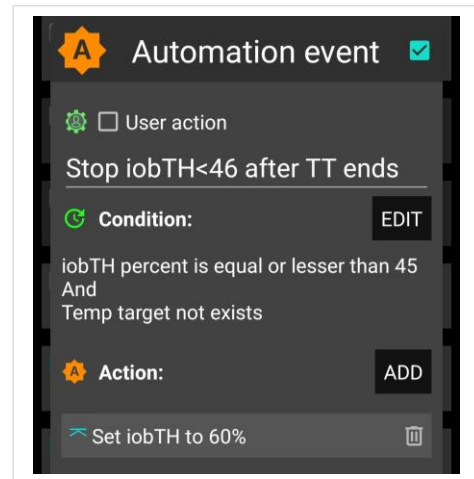
301

302 This feature comes with yet another little tuning opportunity, in which you study your body's
303 response to light exercise (like walking) or to not moving at all (like desk, couch), and select
304 appropriate settings (in AAPS Preferences/OpenAPS SMB/Activity modifies sensitivity/ -> set **two**
305 **scaling factors**).

306 This **will automatically adjust insulin delivery** (basal, ISF, and iobTH; see 1st screen of AAPS
307 AUTO ISF tab (example in [section 5.4.5](#))), to suit activity state of the past minutes (up to 1 hour).

308

309 This autoISF feature (new since V.3.0) is much quicker responding than Autosens or dynamicISF
310 to adjust insulin sensitivity to your current „lifestyle state“ (if it is largely (in-)activity related, which
311 often is the case).



For loopers who do not have huge variations in exercise levels (for which they would need the exercise mode) in their everyday lives, the Activity Monitor might fairly much close the gap towards being able to do a 24/7 hands-off FCL.

The Activity monitor is described in more detail on page 9 of the [_autolSF Quick guide](https://github.com/ga-zelle/autolSF) (<https://github.com/ga-zelle/autolSF>) and in [section 6.6](#)

While the Activity Monitor takes automatic care of *light* deviations from an average activity level, exercise enthusiasts, or heavy workers, should make use of the “*heavier tools*” (which automatically supersede (shut off) Activity Monitor: *Exercise mode*, discussed in [section 6.](#))

5.1.6 Pro/con completely hands-off Full Closed Loop

To stay **24/7** in a **completely „hands-off“ FCL** can be a realistic goal with autolSF, if besides meals also some special challenges, as discussed in this [section 5.1](#), were analyzed and could be addressed. Example see in [Case study 5.3](#).

Clearly it depends very much on your lifestyle, and how interested, willing, and capable you are to recognize, deal with, (and in the future avoid?) situations that get you outside of your desired %TIR on occasion.

So, this is also about what %TIR you are aiming at, and can accept, as it averages out for the week, for instance.

Everybody must weigh for her/himself

- how much **upfront effort** to put into the setting up process, for getting it all 100% automatic
- **or whether to take an easier start, with a couple of situations left to take care of when and as they arise in daily life**

Even if a principal capability for a fully automatic “hands-off” running FCL is given, this still means that

- the user should be knowledgeable about what exactly is going on, and
- have a principal capability to „nudge“, or even to completely take over, in a manual mode.

In the sections that immediately follow, we present the options to nudge or temporarily take over from the AAPS home screen which will be serving as your **FCL cockpit**:

[Section 5.2](#) describes how you can use available “buttons” from your AAPS home screen to **manually** “nudge” aggressiveness of your FCL.

348 5.2 Modulating aggressiveness **manually**, from your DIY-FCL-Cockpit

349

350 Keep in mind:

351 1) The goal should be to **interfere with the loop as little as possible**

352 2) Avoid technical failures and BlueTooth disconnections lasting many minutes (e.g. by leaving
353 your phone in another room for too long).

354 The required insulin would still be supplied *after* BT reconnects. However, the delay can be much more
355 of an issue in FCL than it had been in HCL, where you pre-bolussed.

356

357 In the preceding [sections 4. and 5.1](#) we learned that, under certain conditions, your FCL can run
358 **fully automatically**, without any user interaction.

359

360 However: *Other disturbances* might come up, that **influence** your **temp. insulin sensitivity** dra-
361 matically, and

- 362 • are not noticeable in-time, or foreseeable, by the loop (*e.g. your plan to start exercise in an*
363 *hour or two*), but therefore require temporary modified settings in order to remain in-
364 range, and/or
- 365 • **require** a different “starting point” regarding iob and bg, which translates into **a different**
366 **iobTH** that should **temporarily** be set much lower (*in case of exercise*) or noticeably higher
367 (*e.g. with very fast absorbing carbs in a sweet snack “sin”*) .

368

369 *In these scenarios*, you must find an easy way to

- 370 • manually tweak a setting or two, to temporarily adjust the aggressiveness, or:
- 371 • call up, via a button-push, a pre-programmed routine for automatic management with
372 adjusted aggressiveness.
- 373 • There may also arise a desire to just exit the FCL mode, and “be your own captain” for
374 mastering a special situation.

375

376 5.2.1 Status recognition

377

378 For peace of mind, to learn, and to stay informed (especially in your initial tuning phase, or when
379 your glucose curve goes in unexpected ways) you must be able to find the key parameters that
380 frame and drive the recent and upcoming loop decisions.

381

382 All this is facilitated within seconds right from the AAPS home screen, serving as a **FCL cockpit**

383 Before considering any manual interventions into the ongoing FCL, you should be aware of

- 384 • the prediction curves (bg, and insulin activity), and
- 385 • what the current mode of action is (refer to [section 5.3](#)).

386

387 On that basis you might be able to “nudge” your loop in order to adjust to the disturbance that you

388 see coming up:

389

390 5.2.2 Manual interventions from the (DIY-) FCL cockpit

391

392 5.2.2.1 Temporary tuning of FCL aggressiveness via the top button row on AAPS Home screen

393 (temp. %profile; exercise; TT settings)

394

395 %profile set

396

397 The set **% profile** multiplies with both, the ISF resulting from autoISF, and also with the default

398 iobTH you have set, so both are nicely modulated in a linear way with the % temporarily chosen

399

400 Just taking profile e.g. to 110% for an afternoon might be an easy way to explore whether
401 you might benefit from 10% more “aggressiveness” in your core settings for lunches

402

403 Make sure, though, that your safety limits ([section 2](#)) are set wide enough so they will not
404 cut the extra 10% away!

405

406 Temporary bg target (TT)

407

408 A **lowered** (relative to profile glucose target) temporary **bg target (TT)** signals lowered sensitivity
409 (more insulin need), e.g. as an “EatingSoonTT”.

410

411 An **elevated** TT (e.g. used with exercise) increases sensitivity and hence works in the direction of
412 reducing insulin given by the loop.

413

414 **Caution:** In preferences/SMB..., make sure you set “High TT raises...” and “Low TT
415 lowers sensitivity”.

416 However, advice is NOT to activate the other two options there, that would automatically change TT depending
417 on detected sensitivity. One reason: With our important even/odd TT feature (regarding SMBs, see next
418 headline), we would hate to see TT randomly set to even or odd values.

419

420

421

422 Exercise button

423

424 Moreover, the **exercise button** (top center on your AAPS home screen) can be activated (turns
425 yellow, then). This can ***) further boost** how your set TT elevates the resulting ISF, and sharply
426 lowers iobTH, as often desired for sports. ***) see below in [section 5.3.2](#), and more in [section 6](#)**).

427

428 Even or odd numbered temp. bg target (TT) (of similar level)

429

430 As we learned in [section 5.1.3](#) already, switching between an even or odd bg target can be used to
431 strongly modify loop aggressiveness by switching SMBs on/off temporarily.

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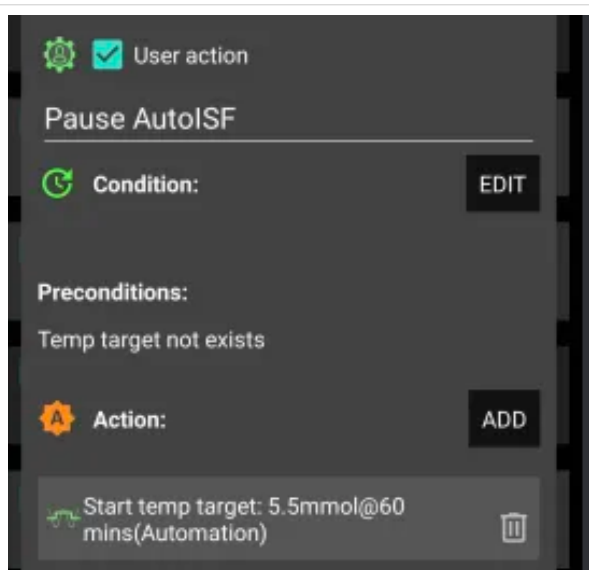
451

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455



It is easy to use the **top right “TT” button** in the AAPS screen to switch from an even target to an odd numbered one, in order to shut off SMBs for a time you can set.

<- An even easier to operate alternative is setting up this **User Action Automation** called “Pause autoISF” (suggested by Slip, Discord Dec.2024).

(Actually, autoISF keeps running, but *it seems paused* because it can’t translate big calculated insulin requirements into SMBs)

446 5.2.2.2 Making temporary changes in settings made in AAPS/preferences/Open APS SMB

447

448 **Caution:** Trouble with some of these is, not to forget to stop and set them back, manually, too.

449

450 Going into AAPS/preferences/Open APS SMB allows to:

- 451 - set milder or stronger ..._ISF_weights
- 452 - set different iob_threshold_percent (or iobMAX)
- 453 - elevate or lower the SMB_delivery_ratio
- 454 - limit or expand max. allowed SMB size
- 455 - change the the even <-> odd logic for SMB on/off

456 Doing temporary changes *in AAPS/preferences* **should be the exception because**

- 457 - they require multiple steps, including entering a password
- 458 - you will often forget to set everything *back to original* settings, a couple of hours (or already
- 459 minutes) later.

460

461 5.2.2.3 Triggered Automations: Grey extra DIY cockpit buttons for pre-programmed “responses”.

462

463 In AAPS, you can build a couple of DIY cockpit features via “User Action Automations”, as
464 described below and in [case studies 5.2](#) and [6.2](#):

465

466 **Snack example.**

467

468 Recognizing conditions for **fully automatic** handling by the loop may not be not possible, or come
469 too late for the loop to act on. Examples would be

470

- 471 • *exercise*: Minimum an hour *before* starting exercise, “the loop should know” to be able to
472 lower iob and elevate bg by the time exercise starts. (For more, see [section 6](#).)

473

- 474 • *snacks*: High carb snacks, sweets, consuming ice cream or having a sweet drink comes
475 with the problem of even steeper glucose rises, but overall a lesser insulin need, compared
476 to major meals (for which we tuned our FCL according to [section 4](#)).

477

478 This not necessarily implies that snacks *need* different settings than a meal. After all, autoISF
479 was designed to react to all available data, especially to where the developing glucose curve is
480 headed. So, depending on your effort to set parameters for a broad variety of meals (notably:
481 how well you avoid to invariably bounce fast against your iobTH), you might be able to accom-
482 modate low carb, snack, and major meals with *one* set of settings.

483

484 In FCL autoISF, this is a bit more difficult than in HCL autoISF applications, because FCL
485 involves revving up iob supply (largely via big bgAccel_ISF-weights) sometimes too much,
486 to be balanced by just a snack getting absorbed.

487

488 **Caution: In case a snack *did* trigger a “full meal response”:**

489 (1) You probably must continue snacking to prevent a hypo from your initial FCL over-reaction.

490 (2) For future days, analyze your data (and snacking habit) to define how to prevent this from happening
491 often.

492

493 For increased comfort and safety, you might have to differentiate, and make use of what follows for
494 the *sweet snack* example, [case study 5.2](#).

495

496 Note that in the iPhone versions of autoISF (**Trio** and **iAPS**) there are no Automations . Instead you
497 need so-called **Middleware**, like for instance suggested for %sensitivity (profile ISF) adaptation by
498 one user here: <https://discord.com/channels/953929437894803478/1025731124615458848/1238099464531611668>
499

500

501 **Tuning aggressiveness**

502

503 A sweet snack likely benefits from even more aggressive initial FCL performance than set for the
504 meals in your normal spectrum of diets.

505

506 Therefore, you could set

- 507 • a higher **temp. profile%** and/or
- 508 • a temp.elevated **bgAccel_ISF-weight** (see screenshot of my Automation).
- 509 • a **low temp. target** (76 mg/dl for instance; this additionally helps maximize the first SMBs
510 that will automatically be triggered at detection of acceleration)..

511

512 When first defining and testing this Automation, also check:

- 513 • that the safety limits as discussed in [section 2](#) will not block the intended elevated aggressiveness
514
- 515 • SMBs will not get outrageously big, and iobTH sometimes exceeded by too much.

516 Note that “the last SMB” is allowed to overshoot the effective iobTH by up to 30%, where it
517 will be cut:-

518

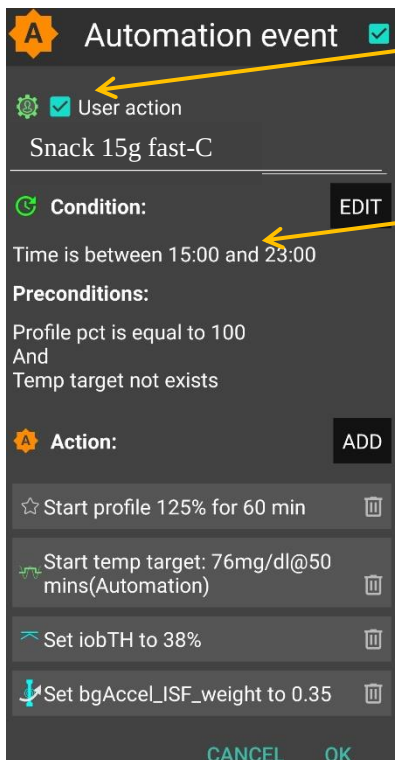
519 **Limiting iob**

520

521 For “just a snack”, total insulin need will be lower than for a meal.

522 If you would just have your sweet drink, and your meal-oriented FCL would “attack”, iob
523 likely would become too high, and a glucose rollercoaster would start, with you needing to
524 consume more =>

525 If you just have a snack, or drink a small glass of juice, you can lower the **iobTH_percent**
526 accordingly.



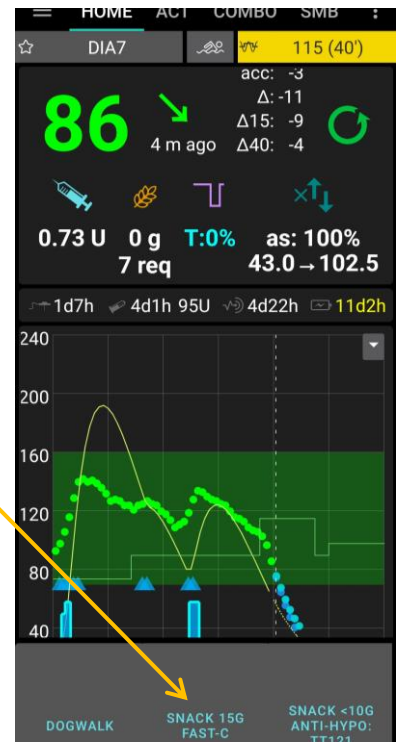
„User action“ is always ticked-on

This will, in the defined time space ..

..offer the “DIY cockpit” button..

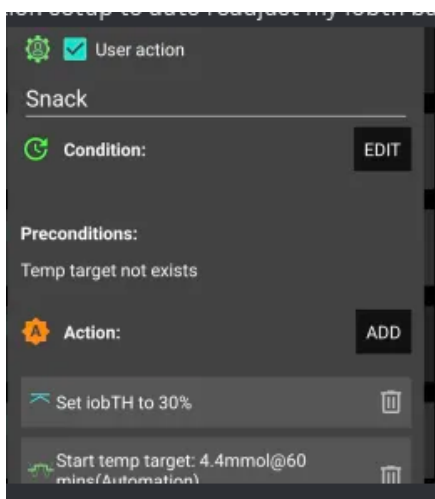
..which I must press any time (~90...30 minutes) before my snack.

Note: Make sure that iobTH and bgAccel_ISF_weight revert to default afterwards



527

528



<.- A simplified way was proposed in Discord (Slip, Dec.2024)

Again, note: Make sure that iobTH% reverts to default afterwards, ideally automatically:

- see Automation event with Action “set iobTH (to default)” (further below)

529

530

531 So, you as part of your setting-up project, you need to research your snack habits (if any),
 532 and over time find out which settings in the snack-related Automations work well. Then, **in**
 533 **everyday life**, you **just press the related button in your cockpit** (which is not time
 534 critical at all, except it should be clicked **latest** a couple of minutes *after* you took the drink
 535 or snack, when acceleration of your bg curve sets in.)

536

537 If you consume more, and also eat something with your sweet drink, this will more
 538 resemble a full meal (... probably with unusual amounts of fast carbs => it is good to use
 539 initially the more aggressive ISF weights as in your snack Automation).

540

541 **Caution:** Pressing your snack button a *second time* would **not** help because the lowered
542 iobTH does not allow iob going high enough. So you are better off just letting your *normal*
543 FCL meal routine run, after your snack mode expired.

544

545 Other options (when you just can't stop snacking) would require a manual modulation
546 regarding %profile and/or bgAccel_ISF, but keeping the full default set iobTH_percent, or even
547 elevating it (refer to [section 5.2.3](#)).

548 If that happens often, define for yourself an *extra* User action Automation for a *bigger* snack (=
549 another grey DIY cockpit button).

550

551 **Caution:** Setting a different iobTH or bgAccel_ISF _weight can not be done with a duration
552 attached.

553

554 Hence you **must** define a suitable **additional**
555 **Automation, that** must be active in tandem, and
556 **restores the iobTH or bgAccel-ISF_weight** in
557 AAPS/Preferences.

558

559 Else, once your Automation set in, it will *forever* shift
560 these important parameter settings!

561

562 Tandem Automation example, for restoring settings in
563 /preferences : ----->

564

565 If for instance you have several Automations that, in combination with a set elevated TT also set a lower
566 iobTH: Don't be fooled, the duration only applies to the TT. You need an extra Automation for all of them.
567 In my Tandem Automation example given above, I picked out the *highest* of the altered iobTH values that my
568 iobTH-reducing Automations can set (45_percent), and then I can automatically restore my default desired
569 60% via this one Automation (see screenshot given above)

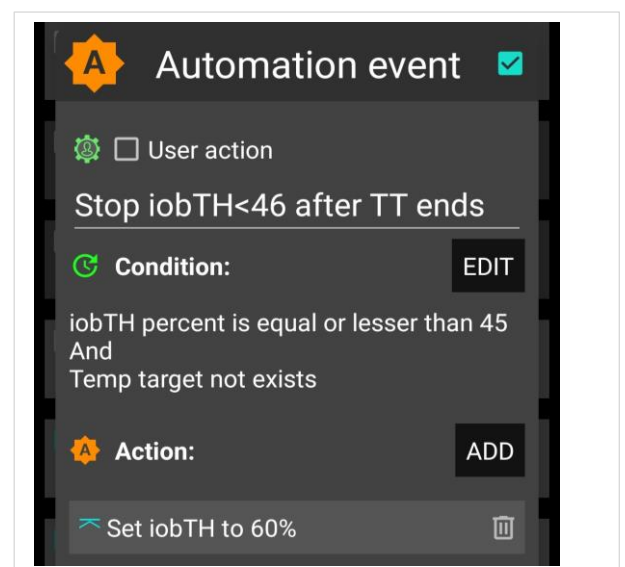
570 In the related Automation, just keep the "User action" box clicked at all times, and define in the
571 Conditions when you want to see that button available for cockpit use (see screenshot above) =>
572 you will see that button offered.

573

574 [Installing your DIY cockpit buttons](#)

575

576 Besides snacks, also any other **recurring special situations can be addressed via a DIY**
577 **cockpit button, and receive different aggressiveness** up to a suitable iobTH level.



578 Over time you can have a big number of User action Automations. Best keep some “shelved”
579 (clicked in-active, top left box) in your long list of potential Automations. Even when active, they
580 only show in your cockpit (bottom grey field of your AAPS home screen) *in the time slot you*
581 *assigned* as potentially relevant (and if all other conditions are met).

582

583 Summary

584

585 In case you do have a snack habit and ...

- 586 • can not find settings, as in [section 4](#), defined for your meals, that also suit your snacks
- 587 • can not pin a time slot or other Condition to it for programming an Automation response
588 as in [section 5.1.4](#)

589 .. then you minimum need a “snack announcement” for which the extra button in your DIY cockpit
590 provides a time-uncritical 1-button-push solution.

591

592 This could be a good solution for kids in kindergarten, too. Make sure caregivers
593 understand to **use it only once** for a snack! **Using the snack button several times in a**
594 **row would limit iobTH at a too-low level!** Note that continued snacking would require iob
595 as for a meal, and that is what the FCL loop takes care of, fully automatically.

596

597 5.2.2.4 Differentiation of FCL aggressiveness using “Automation States”

598

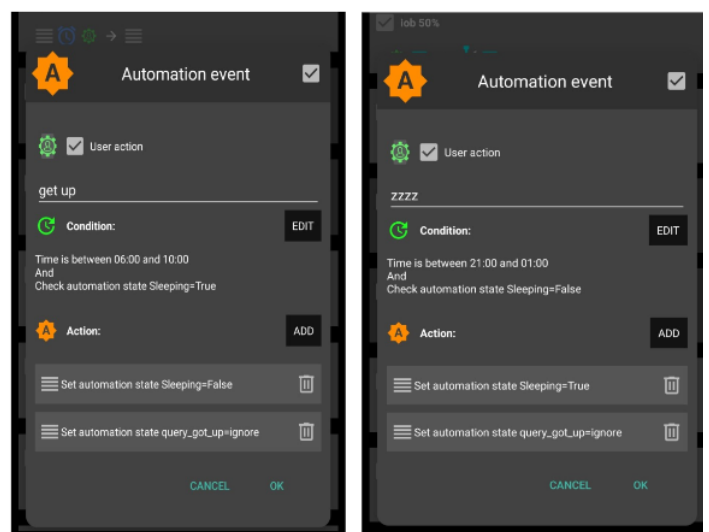
599 This is a prototype implementation in autoISF 3.1.0 that allows further refinement of automating with autoISF
600 parameters.

601 An application
602 example is given in
603 the developer’s
604 autoISF 3.1.0
605 Quick Guide,
606 section “The new
607 Automation of
608 States”:

608 Source:
609 <https://github.com/gazelle/autoISF>

First real use example for disabling inactivity monitoring during sleep

My sleep times are very irregular and I do not want to adapt those hours in the preferences every night. So I created a *state* Sleeping with its two obvious *values*. In addition I created two user action buttons which swap the values for going to bed and getting up, respectively. They trigger visibility in the morning or evening based on likely time windows and *state values*. The real trick now is to interact with those *states* inside the AAPS code rather than just by automations. This offers more complex behaviour. The slight downside is that in this case you have to stick to the names of *state* and *values* shown here. This will take precedence over the sleep hours and its enabling switch. If you do not want this new feature just do not create the *state* Sleeping – or misspell/rename it for your own use.



610

611 5.2.3 Temporarily exiting the FCL

612
613 The “last resort” alternative always is to **temporarily** leave the FCL mode, and handle any
614 disturbance “the traditional way” in **hybrid closed loop**.

615
616 Do not forget that, before any meal starts, giving a bolus will then be necessary again.

617
618 “Enable ISF adaptation..” OFF”

619
620 This switches *all* automatic aggressive adaptations of ISF to the bg curve OFF that autoISF
621 provides.

622
623 To exit the FCL mode, go to AAPS/preferences/*put in your password *)*/OpenAPS SMB/scroll down
624 to autoISF settings and switch “**Enable ISF adaptation..**” OFF”
625 (or, alternatively, set bgAccel_ISF_weight to zero).

626 **) please set a (very short) password, just to avoid accidental clicks when scrolling*

627
628 Set bgAccel_ISF_weight temp. to zero

629
630 If in hybrid closed loop you like e.g. the dura_ISF adaption still, you alternatively could elect to just
631 set bgAccel_ISF_weight temp. to zero.

632
633 Switching back into autoISF FCL

634
635 **Caution:** Unfortunately, there is no way yet for your full closed loop with ISF adaptations to come
636 automatically back on, after a selected time for instance. So **do not forget to switch your**
637 **autoISF fully back on, later.**

638
639 Alternative: Temp. FCL OFF via a new cockpit button

640
641 As this switching back will often be forgotten, an alternative strategy for FCL pauses would involve
642 a custom “User action” Automation, that can give you a “Pause FCL” grey button on your AAPS
643 Home screen (see end of [section 5.2.2.1](#), above).

644
645 Caution though, there is very limited experience with this brand new feature. Make sure your Automation definition really
646 applies a duration (or other condition) that will automatically terminate all settings changes it made. As we have seen e.g.
647 in [section 5.1.4](#), this is not always the case.

648
649

650 Recognizing loop state

651
652 To recognize whether autoISF currently runs with **ISF adaptation** or not, please consult the
653 “**ai: %**” **indicator** below the Autosens% **on the AAPS Home screen**.

654
655 The 1st screen of the **AUTO ISF tab** is probably the best resource at your fingertips to find out in
656 detail what is currently going on.
657 More see in [section 5.3](#) that follows.

658
659 **5.3 Recognizing your loop state in the AAPS home screen**

660
661 **5.3.1 Modulated loop aggressiveness via the 3 top buttons in the AAPS home screen**

662
663 The 3 top buttons (%profile; exercise; TT) of your AAPS Home screen allow temporary modulation
664 of aggressiveness (sensitivity), as may be needed e.g. when

- 665
- 666 • going into meals (EatingSoonTT => providing lower bg “starting point” and more pos. iob),
 - 667 • or doing exercise (Exercise mode, limiting basal, iob etc for hypo prevention).

668
669 **All buttons grey (GGG)** indicates: The **loop is running using** un-modulated **profile** parameters
670

671 Note that autoISF builds on profile (or on a temp. modulated profile) for further “5-
672 minute” modifications (See calculation cascade for used sensitivity (sens) in the AUTO ISF
673 tab. Example see below in [section 5.3.6](#)).

674 Any **yellow** (y or Y) button indicates you are running with a modified (*e*levated, or **l**owered) **profile**
675 **sensitivity**.

676 *Actually, the middle button could be yellow without being active, see table below*

677

	Less aggressive loop = more sensitive user needs temp. less insulin	Standard loop using the set profile	More aggressive loop = less sensitive user needs temp. more insulin
% profile	under 100 % (y)	= 100% (G)	above 100% (Y)
Exercise button	ON (y) Note that it will only work in combination with elevated TT	OFF (G) <i>If ON (y), it is <u>not</u> active but ready to automatically become active whenever any TT above target is set (e.g. by any Automation)</i>	OFF (G)

TT	above profile target (y)	= profile bg target (G)	below profile target (Y)
Example	Exercise mode with weakened %profile, yyy		EatingSoonTT, GGY
..used e.g. in :	Case study 6.2 (yyy)		Case study 6.2 (G_Y via Automation) Case study 5.2 (YGY)

[Case study 6.2](#) uses extremes from both ends of this sensitivity modulation spectrum:

- yyy for reducing basal and iob, and getting weaker ISF, during exercise
- GGY for temporary "EatingSoonTT" boost when, in FCL, a high carb meal had
- The even stronger YGY case is contained in [Case study 5.2](#), where for a high carb snack an extra boost, using >100% profile sensitivity, is temp. added

5.3.2 Color scheme of top cockpit buttons tells kind of closed loop that is running

This section is fairly complicated, not fully checked with Devs, and not crucial to understand:

[Skip and continue at 5.3.3](#)

3 Buttons (%profile; exercise; TT), each in 2 states, yellow (for modified), else: grey (G)

Note that yellow could be less (y) or more (Y) aggressive than standard profile

Any yellow setting modulates sensitivity *or attempts to modulate, whenever another required condition may become true (e.g. also via an Automation)*

Note, though, there are EXCEPTIONS (see below, under GGG, and under Gyy) where sensitivity is auto-adjusted without the user, or an Automation, making a different setting on any of the 3 top buttons. Autosens, or the Activity Monitor, can adjust the %profile sens without the top left button turning yellow or showing that %value. (See sens field in AAPS main screen, or AUTO ISF tab!)

- GGG is the loop running with set profile.

EXCEPTION: If the Activity Monitor or Autosens are running, profile sensitivity could be adjusted to (in)activity despite the %profile button remaining grey (see top of AUTO ISF tab text from AAPS main screen).

Likewise, Autosens can temp. tweak the set profile, with the button still remaining grey,

Overall, 2 exp 3 = **eight principal FCL modes** are possible. They need to be **differentiated further**, based on whether the modulation...:

- ...goes into the **more aggressive** (Y, higher % and/or lower TT), or...
- ...the *less aggressive* (y, lower % and/or higher TT and evtl exercise) direction.
- Note the **Exercise button** works only one way, to make the loop less aggressive ("y"). However, whether the dynamic Exercise Mode (which

709 strongly can adjust sensitivity, see [section 6.1.3.1](#)) is at work, cannot be
 710 recognized by the color of the Exercise button:

Dynamic Exercise Mode (dyn.EM) ?	bg target as in profile: grey	TT > profile target set (Yellow)
Exercise : grey	dyn.EM off	dyn.EM off
Exercise: yellow	dyn.EM <u>off (!) but</u> "on in waiting" = <u>after</u> an Automation sets a TT>profile target	dyn.EM on

711
 712 DEV idea: Should we make the Y to violet or orange (more aggressive) and keep the y yellow (less aggressive
 713 than profile)? Then beginners would easier recognize whether they go more or less aggressive ...

Color combinations	Lower aggressiveness		Higher aggressiveness
GGG		profile	
Gyy GyY	dynamic exercise mode (if TT > profile target)		EatingSoonTT (as TT < profile target: middle y inactive, GyY = GGY 2)
	<u>EXCEPTION</u> : Profile sensitivity is auto- adjusted in dynamic exercise mode, but this leaves the 1 st button grey (G)		
yGy-yGY- YGy-YGY	„traditional“ exercise mode (if <100% sens AND TT > profile target)	mixed cases 1)	boosted EatingSoonTT (if >100% sens AND TT < profile target)
GyG		GyG = GGG 2)	
yGG YGG	Loop adjusted for period of lower insulin need (e.g. 80% for night after exercise)		Loop adjusted for period of increased insulin need (e.g. 120% during sickness)
	The set %profile might be further adjusted by Activity Monitor or by Autosens		
GGy GGY	Loop running with temp elevated target e.g. for enhanced safety against going low (HypoTT)		EatingSoonTT (if TT < profile target)
yyG YyG	same as yGG	y inactive 2)	same as YGG
yyy - Yyy- yyY- YyY	dynamic exercise mode further modulated by <100% profile	mixed cases 3) yyY = yGY, 2)	Boosted EatingSoonTT (if >100% sens AND TT < profile target)(YyY = YGY, 2)

715 1) **yGY** would be the combination of setting a milder % profile, yet orienting the loop to a lower
716 target. Usually we do not use this combination, but it would be OK for an instance where we
717 want to start the next meal at a low target, but we want our loop to be careful and not rush
718 getting there.

719 **YGy** would be the combination of setting a more aggressive % profile, yet orienting the loop to
720 a higher target. Usually we do not use this combination, but it would be OK if we believe we are
721 in a state of enhanced insulin need, and want to get to a set elevated TT (which could be what
722 we want to go max down to to have room for exercise, or just to be sure not to go into a
723 hypoglycemia if we misjudged the symptoms about enhanced insulin need.

724 2) GyG is the same as GGG, differing only *if* (e.g. an Automation) sets a *TT>profile target*. The
725 GyG setting *only then* provides softer loop response, due to the exercise mode kicking in.

726 Generally, a yellow middle button y works really only in the combination ..yy. Any combination ending yG
727 or yY will work the same as if it were ending GG or GY. Although the yellow middle button does not
728 make a difference in these cases, it might be “justified” by the loop showing its readiness to employ the
729 exercise mode, whenever a TT > profile target comes around, e.g. via an Automation.

730
731 3) **yyY** would essentially working like **yGY** (discussed under 1), because the middle button,
732 whether y or G, makes no difference with an elevated TT.

733 Again, this the combination of setting a milder % profile, yet orienting the loop to a lower target.
734 Usually we do not use this combination, but it would be OK for an instance where we want to
735 start the next meal at a low target, but we want our loop to be careful and not rush getting
736 there.

737 **Yyy** would be the combination of setting a more aggressive % profile, yet orienting the loop to a
738 higher target. As opposed to the **YGy** case (discussed under 1), in **Yyy** the Exercise mode is
739 active and strongly reduces aggressiveness. It is hard to imagine, why this combination this
740 combination of settings could make sense; maybe if a “long waved” state of reduced insulin
741 sensitivity meets a case where normally Gyy would be sought (see there).

742

743 5.3.3 Active Automations (or middleware in Trio, iAPS)

744

745 **Active** Automations can temporarily set different aggressiveness and, in AAPS, will automatically adjust to the
746 color scheme discussed in the previous chapter.

747

748 It is important to be aware, which of your Automations (see listed in your AUTO tab, or via the top
749 left burger menu, in AAPS) is **active to eventually “kick in”**.

750 The ones where you clicked “User Action” should also show as an **extra grey button** on the
751 bottom of your AAPS screen (only in the suitable time-of-day bracket, if you assigned one).
752

753 Note that an **Automation might not be permitted** to change settings if still **another Automation**
754 **is running**.

755 Always consider that. **Try to use short durations** in all your Automations! Note that usually you do not lose
756 anything by that, because, if your trigger-Condition remained, the same Automation will repeatedly trigger.
757

758 For instance, you cannot switch from 130% profile to 110%. Either the 130% times out, or you
759 **need an extra “in-between” Automation that terminates** the 130% under described conditions
760 (example see in [Case study 6.2](#)).

761 This “design” is for a good reason: The assumption here is, that your 1st Automation (the
762 130% in the example) is designed well and runs for a reason. It should either “get finished”
763 when the job might be done (and kick in again, if not), or, in exceptional cases, it should be
764 consciously terminated by another well thought through 2nd Automation (describing the
765 conditions in which you would find that *another* Automation now is more important than
766 “finishing up” the one that was already running). That “in-between” Automation makes the
767 loop return to base profile, which is a signal *to all Automations*, to now check whether any
768 conditions exist, to activate a 3rd Automation. (An example is given in [Case study 6.2](#)).
769

770 Advice: 1) Do some “house cleaning”: Occasionally check which of your Automations might work
771 with shorter durations assigned. Reduce your long list of Automations, or at least de-activate those
772 that will not be needed.

773 In case you have many Automations and a very varied life-style, you could make it an evening
774 routine to have only those Automations activated that you might need next day.

775 2) Try to stay away from Automations that also aim at temp. modifying aggressiveness (especially
776 if triggered by bg level). Often, they will not kick in anyways. And generally, it is not a good idea, to
777 “double up” sub-algorithms for tweaking loop behaviors (“loop inside a loop”).(see also at end of
778 [section 5.1.4](#))
779

780 5.3.4 FCL related indicator fields in the AAPS home screen

781
782 In extra data fields of the AAPS main screen you can always see (not change) the key
783 „aggressiveness“ parameters your loop currently operates with (see also home screen
784 example below):

- 785 • To recognize whether autoISF currently runs with **ISF adaptation** or not, please consult the
786 **“ai: %” indicator** below the Autosens% **on the AAPS home screen**.
- 787 • Details for every loop decision see result/debug section of the **AUTO ISF tab**.

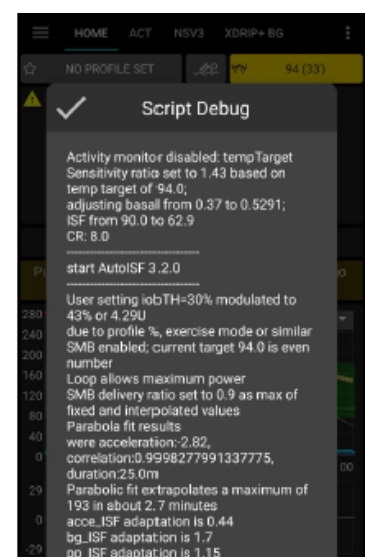
788 5.3.5 Overall home screen:



790 5.3.6 Client screen

791 Apart from the fitted parabola (pink line, around 20:30 in the graph), none of the above new graphics
792 can be shown in AAPSClient (because the related data are not sent to Nightscout. Therefore they cannot be re-
793 loaded to the client from Nightscout. By the way, the same holds for HR and STEPS.)

794 Nevertheless and as an offset, you
795 can now at least request in the client
796 to show the Script Debug from the
797 master because it is so important for
798 AutoISF. For that purpose the field
799 “Algorithm AUTO_ISF” was added
800 besides e.g. Pump and OAPS. It
801 displays the plugin and algorithm
802 active in master. Upon clicking it
803 displays the active Script Debug.
804 Here is an example:



805 5.3.7 autoISF related infos on the main screen

AutoISF specific additions to the home screen

The widget below contains all those additions as it is more or less identical to the top section of the home screen.

1. Time ago since last loop is shown in seconds as long as it is less than 100 seconds ago. Especially with Libre 1 minute values it was more or less frozen at 0m all day.

Meanwhile this feature was added to master AAPS, too.

2. At the bottom right there are two percentages:

as: denotes the Autosense value shown traditionally; **as** is only visible if it deviates from 100% just like in master AAPS

ai: denotes the corresponding sensitivity change based on AutoISF; **ai** is always visible

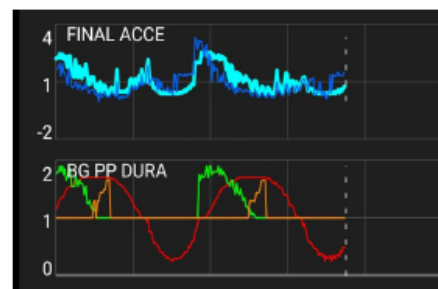


806
807 Note: From autoISF 3.2 onwards, displaying the acc value (top right, underneath the TT field) was discontinued, now that
808 the parabola is shown
809

810 The **extra graphs** available in AAPS below the main glucose chart, recently received an upgrade to include
811 autoISF info:

Showing the AutoISF factors

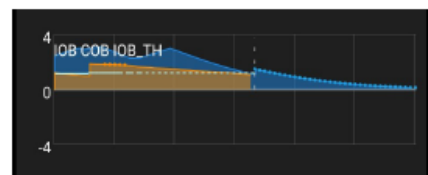
These can be shown as line charts in the small graphs. Within each graph frame the factors are all to the same scale. To add other values makes no sense anyway as they bring their own scale or might even disturb the required unique scaling. To distinguish larger values for acce and final from the other, smaller ones they can be assigned to different graph frames.



Showing iobTH

This can be shown as dotted line. To be exact the effective iobTH is shown as modulated by exercise mode or similar. It makes sense to add IOB and COB to the same small graph to get common scaling between them.

Note: FCL users would have COB=0, i.e. not see an orange field (except as “carb deviation” in an extra graph “DEV”)



812
813
814
815 No more buttons „Insulin“, „Calculator“ etc at bottom of AAPS home screen
816 These buttons are not useful any longer in **FCL**, and should normally be removed (by going into
817 /preferences/overview/buttons and forcing them OFF).
818 The reason why we do this: It really is important to let the loop loop, and not interfere more than
819 absolutely needed. Any bolus the user gives will sure distort the bg curve, on which autoISF,
820 especially when aggressively tuned for FCL, builds a lot of its decisions!
821 *Note:* Eventual attractiveness of doing occasional pre-boli will be discussed in [section 7](#)

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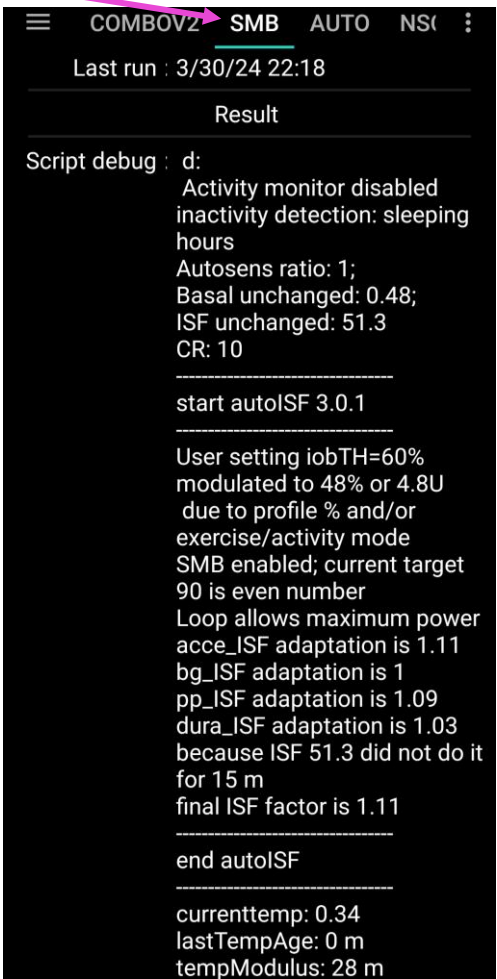
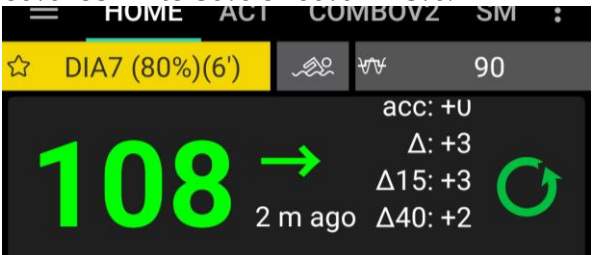
5.3.8 Info given every 5 minutes in the AUTO ISF tab *)

*) in older autoISF versions: "SMB tab"

(If **tab** not shown -> Config.builder/APS: tick box on

When clicking on the AUTO ISF tab, you see how your standard and temporary settings, as well as the latest bg and iob status, influenced the last decision of your FCL.

Example 1: A 80% temp. profile modulates 60% iobTH to 80% of 60% = 48%:



The profile ISF of 41 mg/dl/U got modified by the set 80% temp. profile to $41/0.8 = 51.3$ mg/dl/U, called "ISF unchanged" (before "start autoISF").

autoISF applies the 4 sub-categories (acce, bg, pp and dura_ISF), and depending on the bg curve form suggests various ISF changes.

The final resulting factor "sens" (see flowcharts in [section 3](#)) is 1.11 (in our case, driven by bgAccel_ISF). This changes the 51.3 "unchanged" ISF to $51.3/1.11 = 42.4$ mg/dl/U

Further down in the AUTO ISF tab, you can see how this ISF is applied to define the SMB size to be given, and whether any limitations – notably by autoISFmax, max possible SMB size, or maxIOB – cut the amount.

Message	Condition	What does it affect?
Loop allows maximum power	even target < 100	increase in bg limited to 30%, otherwise no SMB; actual SMB delivery ratio is max of fixed smb_delivery_ratio and linearly growing ratio
Loop allows medium power	even target >= 100	increase in bg limited to 20%, the AAPS default, otherwise no SMB; actual SMB delivery ratio is either fixed smb_delivery_ratio or linearly growing ratio
Loop allows minimal power	odd target	no SMB, only TBR available for action
Loop power level temporarily capped	IOB > effective iobTH	Last SMB was capped to stay below iob threshold + 30% overrun; user defined iobTH potentially modulated by exercise mode, activity monitor and profile percent
Loop allows AAPS power level	no even/odd target option active	SMB enabled/disabled according to standard AAPS rules and settings; no iobTH threshold is active

Note that in the AUTO ISF tab you can only capture and analyze 1 loop decision.

(Occasionally make screenshots ~ every 5 minutes in the critical stages, for analysis of a meal)

850 5.3.9 Info about last 15 autoISF decisions

851
852 Refer to [section 11.2.2](#) for an option that enables extended analysis of the on-going ISF modul-
853 ations from autoISF. (To do this on your **Android** loop phone requires QPython and a logfile
854 emulator).

855
856 You get tables like this ...

17:05

857
858 ... which gives you a quick orientation about recent loop decisions, and relative contributions of the
859 various autoISF contributors (ace, pp, bg, dura).

860
861 A similar table is available also on the **iPhone** for Trio and iAPS users of their autoISF variants.

863 5.3.10 AUTO ISF tab info when operating in 1-minute mode with Libre3

864
865 Users: Please supply text and screenshot

869 **Summary: Your personal FCL cockpit** for maneuvering through disturbances

870
871 A lot of avenues were shown that could help you or your loop maneuver through a variety of
872 “disturbances”. You should not have to try out many of them. Find a way to narrow it down to what
873 really helps in **your** everyday T1D management.

874
875 Try to **keep things as simple and clear as possible**.

876
877 Especially, do not pre-maturely rush into setting up Automations as quick over-patches for what
878 you may not like to see. Limit the number of Automations, and further limit which ones from your
879 list are every-day active (vs. switched off, and only ticked active for special days).

880 See also at end of [section 5.1.4](#)